

Cover Letter

July 4, 2026

Editorial Office

Data & Knowledge Engineering

Elsevier

Dear Editor,

We are pleased to submit our manuscript, "Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning", for consideration as a regular article in Data & Knowledge Engineering.

The manuscript presents Structurally-Calibrated Functional Attribution (SC-FMA), a graph-aware method for converting structured intermediate knowledge objects into auditable priority records. Its novelty is to treat functional attribution as a representation for knowledge audit: local utility or annotation-fidelity signals are preserved while structural dependency, redundancy, and bottleneck roles are recorded as curation fields.

The manuscript fits Data & Knowledge Engineering because it addresses structured audit prioritization for knowledge-intensive workflows, aligning with the journal's interests in knowledge representation, knowledge-base maintenance, and management of engineered data and knowledge artifacts. SC-FMA turns traces of retrieval checks, entity bindings, rule-like steps, and process annotations into graph-aware audit records for fixed-budget curation.

The evidence package includes locked PRM800K annotation evidence where `w_struct` is the primary real-data signal and Ridge preserves it while exposing a decomposable audit record, a Countries-KG representation study showing that typed ontology edges can change structural-dependency allocation, rule-derived audit-target retrieval, and controlled synthetic calibration.

The manuscript keeps deployment, downstream-training, external-transfer, and causal claims separate from the reported audit-prioritization evidence.

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Sincerely,

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